Department of Mathematics
IIT Hyderabad
MA1130: Vector Calculus

Credits: 1

Fall 2026

October 26, 2026

Time: 30 Minutes | *Quiz* | Maximum Marks: 10

Name: Roll No:

Instructions:

1. This question paper contains **4 questions** in total, printed on 4 **pages**. Each question carries **2.5 marks**.
2. Mobile phones and electronic gadgets are strictly prohibited during the examination. Any violation of these rules, including using or possessing such devices, will result in a **-10 marks**, with no further discussion or consideration.

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1. Find the directional derivative of $f(x, y, z) = \sin(x^2yz)$ at the point $(1, 2, 1)$ in the direction of the vector $(2, -1, 2)$.

2. Find the volume V of the solid bounded by the three coordinate planes and the plane $3x + 2y + 4z = 12$.

3. Find the volume under the plane $z = 12x + 8y$ over the region $R = \{(x, y) : 0 \leq x \leq 1,\ 0 \leq y \leq 3x^3\}$.

4. For a rectangular solid of volume 1000 cubic meters, find the dimensions that will minimize the total surface area, subject to the condition that the length is twice the width.